

Solubility Model

Flow Equations for Simplified Henry's Law Flash for Species i

$$\begin{aligned}\psi_i \equiv & \Delta \left[T_o^m y_{io}^m (\Delta p^{n+1} - \gamma_o^m \Delta D) \right] + \Delta \left[T_g^m y_{ig}^m (\Delta p^{n+1} + \Delta p_{cog}^m - \gamma_g^m \Delta D) \right] \\ & + \Delta \left[T_a^m y_{ia}^m (\Delta p^{n+1} - \Delta p_{cao}^m - \gamma_a^m \Delta D) \right] + q_i^{n+1} \\ - \frac{V}{\Delta t} & \left[\phi^{n+1} (N_i^{n+1} + N_{ai}^{n+1}) - \phi^n (N_i^n + N_{ai}^n) \right] = 0\end{aligned}\tag{A5.1}$$

For Water

$$\begin{aligned}\psi_{n_c+1} = & \Delta \left[T_a^m y_{wa}^m (\Delta p^{n+1} - \Delta p_{cao}^m \gamma_a^m \Delta D) \right] + q_{n_c+1}^{n+1} \\ - \frac{V}{\Delta t} & \left[\phi^{n+1} N_{aw}^{n+1} - \phi^n N_{aw}^n \right] = 0\end{aligned}\tag{A5.2}$$

Volume Balance Equation

$$\psi_v = V \frac{\sum_{i=1}^{n_c} (N_i^{n+1} + N_{ai}^{n+1}) + N_{aw}^{n+1}}{(\rho_o S_o + \rho_g S_g + \rho_a S_a)^{n+1}} - V \phi^{n+1} = 0\tag{A5.3}$$

where

V = grid block volume

T_j = transmissibility of phase j , $j = o, g, a$

y_{ij} = mole fraction of component i in phase j

$i = 1, \dots, n_c$ for oil and gas phases

$i = 1, \dots, n_c+1$ for aqueous phase

γ = gravity potential of phase j

ΔD = depth in gravity direction

N_i = moles of species i in the oil and gas phases

N_{ai} = moles of species i in the aqueous phase

Relationship Between N_i and N_{ai}

Perform a two phase flash and calculate f_o or f_g or both

$$y_{ai} = f_{gi} / H_i \text{ from Henry's law where } H_i \text{ is the Henry's law constant} \quad (\text{A5.4})$$

$$y_{aw} = 1 - \sum_{i=1}^{n_c} y_{ai} \quad (\text{A5.5})$$

N_{aw} is a primary variable and is thus a known quantity

$$N_{ai} = y_{ai} \left(\frac{N_{aw}}{y_{aw}} \right) \quad (\text{A5.6})$$

Total moles = $N_{Ti} = N_{ai} + N_i$ where N_i are the combined moles in the oil and gas phases

In Equation [\(A5.4\)](#) above the Henry's law constant H_i is determined from the following equation

$$H_i = H_i^* \exp \left[\frac{v_i^\infty (p - p_{ref})}{RT} \right] \quad (\text{A5.7})$$

where H_i^* is the reference Henry's law constant at a pressure p_{ref} , v_i^∞ is the partial molar volume of component i at infinite dilution. R is the universal gas constant and T is the temperature.